



Introduction to Nextclade

Understanding virus sequences made simple

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What is Nextclade?

A Tool for Understanding Viral Evolution

Key Functionalities:

- Analyzing viral genomes
- Full functionality in web browser & also available as CLI
- Free and open-source
- Supports important viruses such as SARS-CoV-2, Influenza, RSV, Mpox,...

→ Nextclade provides a fast and user-friendly way to interpret viral sequencing data for research and surveillance.



Why is Nextclade useful to us?

When analyzing a virus sample, Nextclade helps us understand:

- Is the genome sequence accurate and complete?
- Which variant or lineage does this sample belong to?
- How different is the sample from the reference?
- How is it related to other known virus samples?

Why this matters:

Nextclade makes it easier to quickly check, compare and classify viral genomes



How does Nextclade work? (Simplified)

Input:

- Your sequence
- Nexclade dataset

1. Quality Check

- Check for errors (N's)
- Flag suspicious mutations

2. Compare & Align

- Line up with reference

3. Find mutations

- Spot changes
(A → T, C → G, ...)

4. Assign to Variant

- Match to known variants

5. Place on family tree

- Show where it belong
- Find closest relatives

Results Report



Advantages of using Nextclade

Privacy First

- All processing happens on your computer & your data never leaves the browser (no uploads to external servers)

Fast & Accessible

- Analyze hundreds of sequences in minutes
- No installation required (in web version)

Built-In Quality Control:

- Automatically flags problematic sequences & color-coded warning
- Helps catch errors early



Understanding the Output

1. Quality Score: Overall quality

 **Green** = Good sequence

 **Yellow** = Some issues

 **Red** = Poor quality, use with caution

Individual QC Rules

■ Missing data (N)

■ Mixed sites (M)

■ Private mutations (P)

■ Mutation clusters (C)

■ Stop codons (S)

■ Frame shifts (F)

- 
- Low coverage
 - Low variant QC
 - Variant ambiguity
 - ...

<https://docs.nextstrain.org/projects/nextclade/en/stable/user/algorithm/06-quality-control.html#>



Understanding the Output

2. Variant/Clade Assignment

- Helps track which variants are circulating
- Allows you to compare samples from different regions and time periods
- „Your sequence is BA.5.1“

Nextclade						
		Start	Dataset	Results	Tree	Export
				Do		
#	i	Sequence name	QC	Clade	Pango lineage (Nextclade)	WHO name
0	0	✓ OY754687	N M P C F S	23B	XBB.1.16.11	
1	1	✓ USA/CA-LACPHL-AY03266/2023	N M P C F S	23F	EG.5.1.13	
2	4	✓ USA/CA-LACPHL-AY03267/2023	N M P C F S	23F	EG.5.1.16	
3	3	✓ OY754651	N M P C F S	23F	EG.5.1	
4	2	✓ OY754528	N M P C F S	23E	GE.1	
5	5	✓ OY754526	N M P C F S	23E	GJ.2	
6	6	✓ USA/WA-UW-23102346451/2023	N M P C F S	recombinant	XCR	recombinant
7	7	✓ USA/WA-UW-23102330989/2023	N M P C F S	23F	HV.1	
8	8	✓ USA/CA-LACPHL-AY03271/2023	N M P C F S	23G	XBB.1.5.70	
9	10	✓ USA/CA-LACPHL-AY03247/2023	N M P C F S	23B	JF.4	
10	9	✓ OY754632	N M P C F S	23I	JN.3.3	Omicron
11	11	✓ USA/CA-LACPHL-AY03264/2023	N M P C F S	23A	HR.1.1	
12	12	✓ OY754626	N M P C F S	23F	EG.5.1.3	
13	13	✓ USA/CA-LACPHL-AY03246/2023	N M P C F S	23C	DV.7.1.4	Omicron



Understanding the Output

3. Mutation List

- Shows specific changes in the virus
 - **A789G**: “nucleotide change at position 789 from A to G”
 - **S:L614E**: “amino acid change in spike protein position 614 from L to E”

4. Phylogenetic Placement

- Places the sequence into an evolutionary tree
- Shows how closely related it is to other viral samples
- Helps trace the emergence and spread of variants



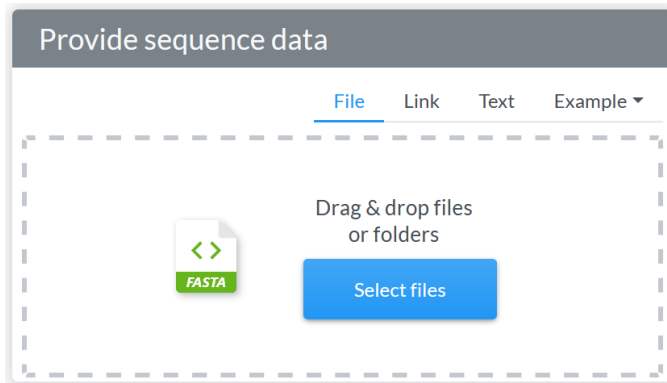
Getting Started

Web version: <https://clades.nextstrain.org>

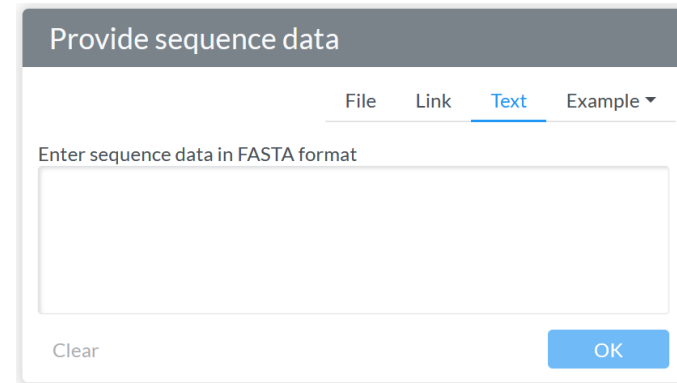
Steps:

1. Upload your sequence file in FASTA format (uncompressed)

Optionally you can also copy & paste the sequence



The screenshot shows the 'Provide sequence data' interface with the 'File' tab selected. It features a dashed box for file upload, a 'FASTA' icon, and a 'Select files' button. The 'Link', 'Text', and 'Example' tabs are also visible.



The screenshot shows the 'Provide sequence data' interface with the 'Text' tab selected. It features a text input area for pasting sequence data, a 'Clear' button, and an 'OK' button. The 'File', 'Link', and 'Example' tabs are also visible.



Getting Started

2. Select reference dataset

- After selecting the sequence, the website will suggest a reference dataset automatically
- You can disable this function and instead choose the dataset freely

Selected reference dataset

Single dataset Multiple datasets

☒ Suggest automatically [Reset](#) [Re-suggest](#)

SARS-CoV-2

Reference: Wuhan-Hu-1/2019 (MN908947)

Updated at: 2026-04-21 09:39:50 (UTC)

Dataset name: nextstrain/sars-cov-2/wuhan-hu-1/orfs

[Open tree](#) [Load example](#)

[Change reference dataset](#) [Run](#)

Select reference dataset

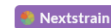
☐ Suggest automatically[Reset](#)[Suggest](#)

SARS-CoV-2

Reference: Wuhan-Hu-1/2019 (MN908947)

Updated at: 2026-04-21 09:39:50 (UTC)

Dataset name: nextstrain/sars-cov-2/wuhan-hu-1/orfs

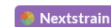


Influenza A H1N1pdm HA

Reference: A/Wisconsin/588/2019 (MW626062)

Updated at: 2026-04-14 11:55:23 (UTC)

Dataset name: nextstrain/flu/h1n1pdm/ha/MW626062





Getting Started

3. Press “Run” and explore the results!

Run

Nextclade Start Dataset Results Tree Export ☒ Done. Total sequences: 165. Succeeded: 165 Settings About Citation Docs CLI EN																
#	i	Sequence name	QC	Clade	Pango lineage (Nextclade)	WHO name	Mut.	non-ACGTN	Ns	Cov.	Gaps	Ins.	FS	SC	Genetic feature	
?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	Gene	Relative to
0	0	✓ OY754687	N M P C F S	23B	XBB.1.16.11		104	2	160	99.5%	56	0	0	0 (1)		
1	1	✓ USA/CA-LACPHL-AY03266/2023	N M P C F S	23F	EG.5.1.13		109	0	1229	95.1%	39	0	0	0 (1)		
2	4	✓ USA/CA-LACPHL-AY03267/2023	N M P C F S	23F	EG.5.1.16		106	0	0	99.3%	30	0	0	0 (1)		
3	3	✓ OY754651	N M P C F S	23F	EG.5.1		103	2	3400	88.6%	9	0	0	0 (1)		
4	2	✓ OY754528	N M P C F S	23E	GE.1		69	14	11269	62.3%	0	0	0	0		
5	5	✓ OY754526	N M P C F S	23E	GJ.2		98	0	1944	93.5%	9	0	0	0		
6	6	✓ USA/WA-UW-23102346451/2023	N M P C F S	recombinant	XCR	recombinant	110	0	0	98.8%	30	0	0	0 (1)		
7	7	✓ USA/WA-UW-23102330989/2023	N M P C F S	23F	HV.1		108	0	0	98.8%	30	0	0	0 (1)		
8	8	✓ USA/CA-LACPHL-AY03271/2023	N M P C F S	23G	XBB.1.5.70		98	0	756	96.7%	30	0	0	0 (1)		
9	10	✓ USA/CA-LACPHL-AY03247/2023	N M P C F S	23B	JF.4		110	0	129	98.8%	32	0	0 (1)	0 (1)		
10	9	✓ OY754632	N M P C F S	23I	JN.3.3	Omicron	114	2	185	99.4%	3	0	0	0		
11	11	✓ USA/CA-LACPHL-AY03264/2023	N M P C F S	23A	HR.1.1		109	0	1	99.2%	30	0	0	0 (1)		
12	12	✓ OY754626	N M P C F S	23F	EG.5.1.3		106	2	143	99.5%	9	0	0	0 (1)		
13	13	✓ USA/CA-LACPHL-AY03246/2023	N M P C F S	23C	DV.7.1.4	Omicron	108	0	1398	94.6%	27	0	0	0		



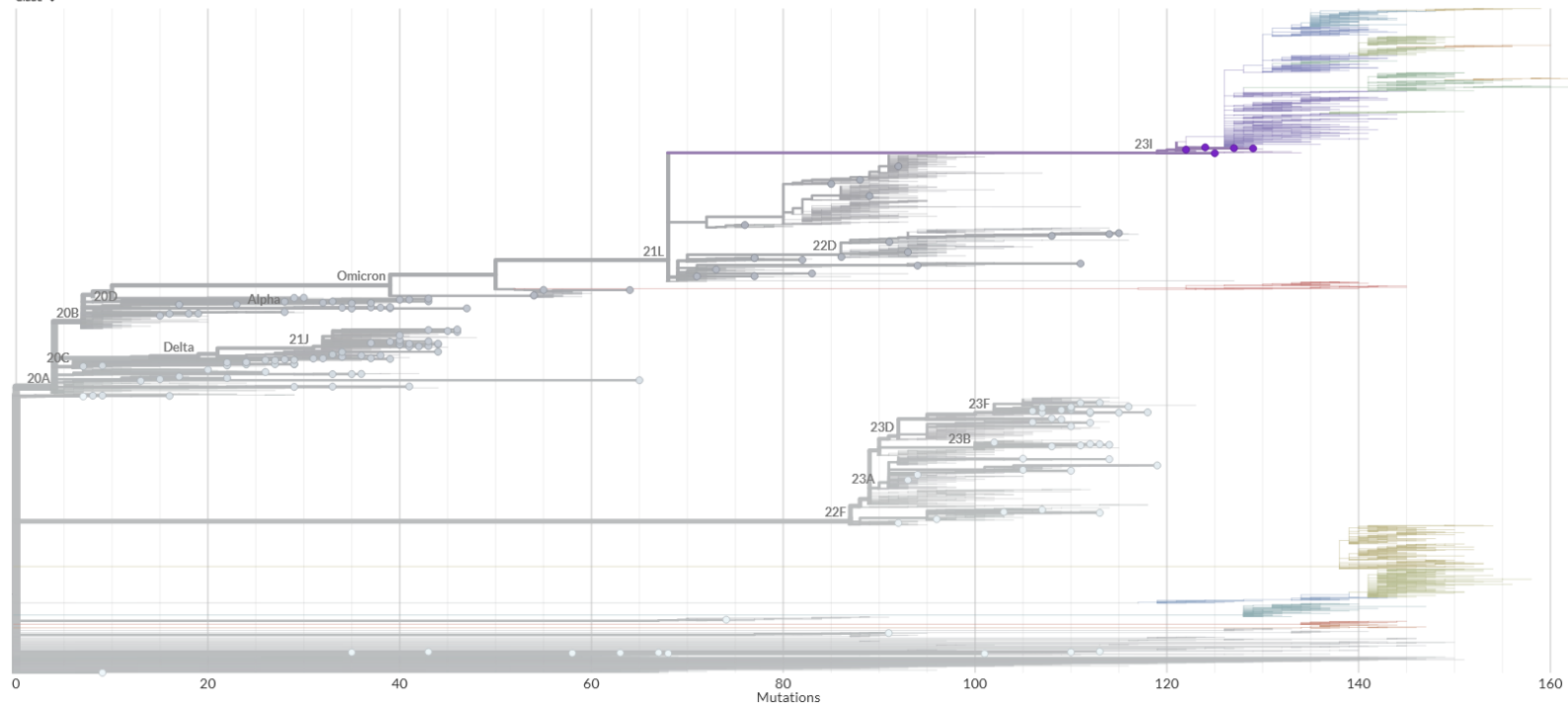
Getting Started

Filtered to [New \(165\)](#)

Phylogeny

Clade ▾

[ZOOM TO SELECTED](#) [ZOOM TO ROOT](#)





Nextclade CLI

Need more power?

- Command-line tool is available for batch processing:
 - Can be integrated into automated pipelines
 - Process thousands of sequences
- → Omnifluss_downstream (https://github.com/rki-mf1/omnifluss_downstream)
 - Finds best dataset, downloads dataset, runs Nextclade, creates a tree

For more information, see:

<https://docs.nextstrain.org/projects/nextclade/en/stable/user/nextclade-cli.html>



Links & useful ressources

Any Questions?

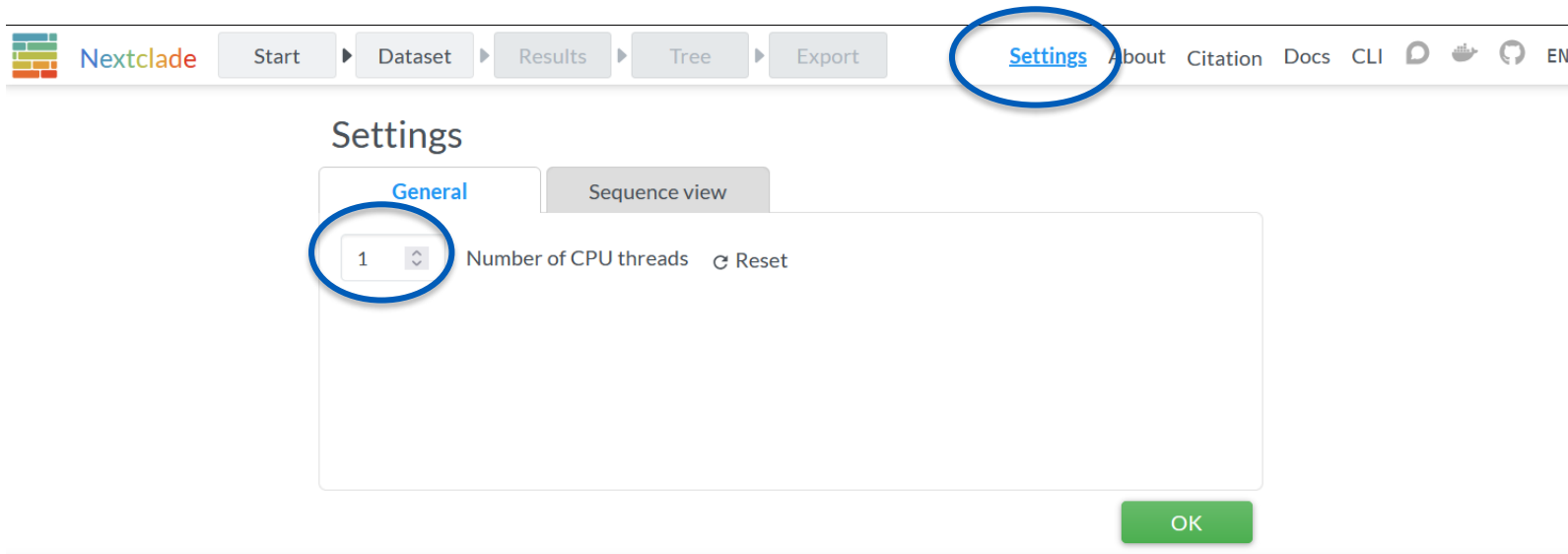
- Website: <https://clades.nextstrain.org>
- Documentation: <https://docs.nextstrain.org>
- GitHub: <https://github.com/nextstrain/nextclade>
- Discussion Forum: <https://discussion.nextstrain.org>

Citation:

Aksamentov et al., (2021). Nextclade: clade assignment, mutation calling and quality control for viral genomes. Journal of Open Source Software, 6(67), 3773, <https://doi.org/10.21105/joss.03773>



Set resources in Nextclade Web



Nextclade Start Dataset Results Tree Export **Settings** About Citation Docs CLI D GitHub EN

Settings

General Sequence view

1 Number of CPU threads [Reset](#)

OK